

Solution Tutorial VII – Design Chart

A modern, three-engine passenger jet is to be developed based on the fundamental aircraft design methods. Determine the following limit curves and present them graphically in a T_0/w_{TO} vs. w_{TO}/S chart. Attention: Do not use the A320 dataset for this tutorial!

Task 1

Take-off distance (ground roll only) of 1400m from an airport at mean sea level altitude assuming that $\bar{T} = T_0$.

Equation to be used (derivation see lecture notes):

$$\frac{T_0}{w_{TO}} \geq \frac{C^2}{C_{L,max,TO} \cdot s_1 \cdot \rho \cdot g} \cdot \frac{w_{TO}}{S} + \frac{C}{2 \cdot C_{L,max,TO}} \left(C_{D0,TO} + \frac{C_{L,TO}^2}{3 \cdot \pi \cdot AR \cdot e} - \mu_f \cdot C_{L,TO} \right) + \mu_f$$

Coefficients of equation:

- $C = 1.15$
- $C_{L,max,TO} = 2$
- $s_1 = 1400 \text{ m}$
- $\rho = 1.225 \text{ kg/m}^3$ (mean sea level)
- $g = 9.81 \text{ m/s}^2$ (gravitational acceleration)
- $C_{D0,TO} = 0.02$
- $C_{L,TO} = 1.152$
- $AR = 10$
- $e = e_{TO} = 0.80$
- $\mu_f = 0.02$ (friction coefficient)

Therefore:

$$\begin{aligned} \frac{T_0}{w_{TO}} &\geq \frac{1.15^2}{2 \cdot 1400 \cdot 1.225 \cdot 9.81} \frac{w_{TO}}{S} + \frac{1.15^2}{2 \cdot 2} \cdot \left(0.02 + \frac{1.152^2}{3 \cdot \pi \cdot 10 \cdot 0.80} - 0.02 \cdot 1.152 \right) + 0.02 \\ &\Rightarrow \frac{T_0}{w_{TO}} \geq 3.93 \cdot 10^{-5} \cdot \frac{w_{TO}}{S} + 0.0248 \end{aligned}$$

Task 2

Minimum prescribed climb gradient in the case of an engine failure (OEI) for trijets at an altitude of $h = 0 \text{ m}$ at $V = 86 \text{ m/s}$ with a maneuvering weight of $w_{man} = 0.990 \cdot w_{TO}$.

Equation to be used (derivation see lecture notes):

$$\frac{T_0}{w_{TO}} \geq \frac{T_0}{T_{man}} \cdot \left[\frac{SEP}{V} \cdot \frac{w_{man}}{w_{TO}} + C_{D0} \cdot \bar{q}_{man} \cdot \frac{1}{\frac{w_{TO}}{S}} + \frac{k}{\bar{q}_{man}} \cdot \left(\frac{w_{man}}{w_{TO}} \right)^2 \cdot \frac{w_{TO}}{S} \right]$$

Coefficients of equation:

- $V = 86 \text{ m/s}$
- $\rho = 1.225 \text{ kg/m}^3$ (mean sea level)
- $\frac{T_0}{T_{man}} = \frac{T_0}{\frac{2}{3} \cdot T_0 \cdot \left(\frac{V}{V_i}\right)^{n_V} \cdot \left(\frac{\rho}{\rho_i}\right)^{n_\rho}} = \frac{\frac{3}{2}}{\left(\frac{V}{V_i}\right)^{n_V} \cdot \left(\frac{\rho}{\rho_i}\right)^{n_\rho}} = \frac{\frac{3}{2}}{\left(\frac{86}{170}\right)^{-0.4} \cdot \left(\frac{1.225}{1.225}\right)^{0.7}} = 1.14$
(value of $\frac{2}{3}$ because only 2 out of 3 engines are running in the OEI case)
- $SEP \approx \gamma \cdot V = 2.7\% \cdot 86 = 2.32 \text{ m/s}$ (γ from CS-25 regulations excerpt for trijets)
- $C_{D0} = 0.02$
- $\bar{q}_{man} = \frac{1}{2} \cdot \rho \cdot V^2 = \frac{1}{2} \cdot 1.225 \cdot 86^2 = 4530 \text{ N/m}^2$
- $C_{L\alpha} = \frac{2 \cdot \pi \cdot AR}{2 + AR} = \frac{2 \cdot \pi \cdot 10}{2 + 10} = 5.24 \text{ /rad}$
- $k = 1/C_{L\alpha} = 1/5.24 = 0.191$ (larger C_L expected: slow flight shortly after take-off)
- $\frac{w_{man}}{w_{TO}} = 0.99$

Therefore:

$$\begin{aligned} \frac{T_0}{w_{TO}} &\geq 1.14 \cdot \left[\frac{2.32}{86} \cdot 0.990 + 0.02 \cdot 4530 \cdot \frac{1}{\frac{w_{TO}}{S}} + \frac{0.191}{4530} \cdot (0.990)^2 \cdot \frac{w_{TO}}{S} \right] \\ \Rightarrow \frac{T_0}{w_{TO}} &\geq 0.0304 + 103 \cdot \frac{1}{\frac{w_{TO}}{S}} + 4.71 \cdot 10^{-5} \cdot \frac{w_{TO}}{S} \end{aligned}$$

Task 3

Service ceiling of $h = 12.5 \text{ km}$ at $M = 0.78$ with $w_{man} = 0.956 \cdot w_{TO}$ (residual climb rate of $SEP = 500 \text{ ft/min}$ necessary).

Equation to be used (derivation see lecture notes):

$$\frac{T_0}{w_{TO}} \geq \frac{T_0}{T_{man}} \cdot \left[\frac{SEP}{V} \cdot \frac{w_{man}}{w_{TO}} + C_{D0} \cdot \bar{q}_{man} \cdot \frac{1}{\frac{w_{TO}}{S}} + \frac{k}{\bar{q}_{man}} \cdot \left(\frac{w_{man}}{w_{TO}} \right)^2 \cdot \frac{w_{TO}}{S} \right]$$

Coefficients of equation:

- $V = M \cdot a = 0.78 \cdot 295 = 230 \text{ m/s}$
- $\rho = 0.287 \text{ kg/m}^3$ (from standard atmosphere)
- $\frac{T_0}{T_{man}} = \frac{T_0}{T_0 \cdot \left(\frac{V}{V_i}\right)^{n_V} \cdot \left(\frac{\rho}{\rho_i}\right)^{n_\rho}} = \frac{1}{\left(\frac{V}{V_i}\right)^{n_V} \cdot \left(\frac{\rho}{\rho_i}\right)^{n_\rho}} = \frac{1}{\left(\frac{230}{170}\right)^{-0.4} \cdot \left(\frac{0.287}{1.225}\right)^{0.7}} = 3.12$
- $SEP = 500 \text{ ft/min} = 2.54 \text{ m/s}$
- $\frac{w_{man}}{w_{TO}} = 0.956$
- $C_{D0} = 0.018$
- $\bar{q}_{man} = \frac{1}{2} \cdot \rho \cdot V^2 = \frac{1}{2} \cdot 0.287 \cdot 230^2 = 7590 \text{ N/m}^2$
- $k = \frac{1}{\pi \cdot AR \cdot e_C} = \frac{1}{\pi \cdot 10 \cdot 0.85} = 0.0374$ (cruise flight, small/moderate C_L assumed)

Therefore:

$$\begin{aligned} \frac{T_0}{w_{TO}} &\geq 3.12 \cdot \left[\frac{2.54}{230} \cdot 0.956 + 0.018 \cdot 7590 \cdot \frac{1}{\frac{w_{TO}}{S}} + \frac{0.0374}{7590} \cdot (0.956)^2 \cdot \frac{w_{TO}}{S} \right] \\ \Rightarrow \frac{T_0}{w_{TO}} &\geq 0.0329 + 426 \cdot \frac{1}{\frac{w_{TO}}{S}} + 1.41 \cdot 10^{-5} \cdot \frac{w_{TO}}{S} \end{aligned}$$

Task 4

Maximum flight Mach number of $M = 0.90$ at an altitude of $h = 10 \text{ km}$ with $w_{man} = 0.850 \cdot w_{TO}$.

Equation to be used (derivation see lecture notes):

$$\frac{T_0}{w_{TO}} \geq \frac{T_0}{T_{man}} \cdot \left[C_{D0} \cdot \bar{q}_{man} \cdot \frac{1}{\frac{w_{TO}}{S}} + \frac{k}{\bar{q}_{man}} \cdot \left(\frac{w_{man}}{w_{TO}} \right)^2 \cdot \frac{w_{TO}}{S} \right]$$

Coefficients of equation:

- $V = M \cdot a = 0.90 \cdot 299.4 = 269 \text{ m/s}$
- $\rho = 0.412 \text{ kg/m}^3$ (from standard atmosphere)
- $\frac{T_0}{T_{man}} = \frac{T_0}{T_0 \cdot \left(\frac{V}{V_i}\right)^{n_V} \cdot \left(\frac{\rho}{\rho_i}\right)^{n_\rho}} = \frac{1}{\left(\frac{V}{V_i}\right)^{n_V} \cdot \left(\frac{\rho}{\rho_i}\right)^{n_\rho}} = \frac{1}{\left(\frac{269}{170}\right)^{-0.4} \cdot \left(\frac{0.412}{1.225}\right)^{0.7}} = 2.58$
- $C_{D0} = 0.019$
- $\bar{q}_{man} = \frac{1}{2} \cdot \rho \cdot V^2 = \frac{1}{2} \cdot 0.412 \cdot 269^2 = 14900 \text{ N/m}^2$
- $k = \frac{1}{\pi \cdot AR \cdot e_C} = \frac{1}{\pi \cdot 10 \cdot 0.85} = 0.0374$ (cruise flight, small/moderate C_L assumed)
- $\frac{w_{man}}{w_{TO}} = 0.85$

Therefore:

$$\begin{aligned} \frac{T_0}{w_{TO}} &\geq 2.58 \cdot \left[0.019 \cdot 14900 \cdot \frac{1}{\frac{w_{TO}}{S}} + \frac{0.0374}{14900} \cdot (0.850)^2 \cdot \frac{w_{TO}}{S} \right] \\ \Rightarrow \frac{T_0}{w_{TO}} &\geq 730 \cdot \frac{1}{\frac{w_{TO}}{S}} + 4.68 \cdot 10^{-6} \cdot \frac{w_{TO}}{S} \end{aligned}$$

Task 5

Vertical gust load factor of $\Delta n_z \leq 1.5$ at an altitude of $h = 4000 \text{ m}$ at $V = 230 \text{ kn}$ in the case of a vertical wind gust with velocity $w_g = 15 \text{ m/s}$ at $w_{man} = 0.750 \cdot w_{TO}$.

Equation to be used (derivation see lecture notes):

$$\frac{w_{TO}}{S} \geq \frac{C_{L\alpha} \cdot \rho \cdot V \cdot w_g}{\Delta n_z \cdot 2 \cdot \frac{w_{man}}{w_{TO}}}$$

Coefficients of equation:

- $C_{L\alpha} = \frac{2 \cdot \pi \cdot AR}{2 + AR} = \frac{2 \cdot \pi \cdot 10}{2 + 10} = 5.24 \text{ /rad}$
- $\rho = 0.819 \text{ kg/m}^3$ (from standard atmosphere)
- $V = 230 \text{ kn} = 230 \text{ kn} \cdot 0.5144 \frac{\text{m/s}}{\text{kn}} = 118 \text{ m/s}$ (convert to m/s)
- $w_g = 15 \text{ m/s}$
- $\Delta n_z = 1.5$
- $\frac{w_{man}}{w_{TO}} = 0.75$

Therefore:

$$\frac{w_{TO}}{S} \geq \frac{5.24 \cdot 0.819 \cdot 118 \cdot 15}{1.5 \cdot 2 \cdot 0.750} = 3380 \text{ N/m}^2$$

Task 6

1-g minimum speed of $V_S = 130 \text{ kn}$ at mean sea level with $w_{man} = 0.75 \cdot w_{TO}$.

Equation:

$$L = 1 \cdot w_{man} \Rightarrow \bar{q}_{man} \cdot S_{ref} \cdot C_{L,max} = w_{man} = \frac{w_{man}}{w_{TO}} \cdot w_{TO}$$

$$\Rightarrow \frac{w_{TO}}{S} \leq \frac{\bar{q}_{man} \cdot C_{L,max}}{\frac{w_{man}}{w_{TO}}}$$

Coefficients of equation:

- $V = 130 \text{ kn} = 130 \text{ kn} \cdot 0.5144 \frac{\text{m/s}}{\text{kn}} = 66.9 \text{ m/s}$
- $\rho = 1.225 \text{ kg/m}^3$ (mean sea level)

- $\bar{q}_{man} = \frac{1}{2} \cdot \rho \cdot V^2 = \frac{1}{2} \cdot 1.225 \cdot 66.9^2 = 2740 \text{ N/m}^2$
- $C_{Lmax} = 2.4$
- $\frac{w_{man}}{w_{TO}} = 0.75$

Therefore:

$$\frac{w_{TO}}{S} \leq \frac{2740 \cdot 2.4}{0.75} = 8770 \text{ N/m}^2$$

Task 7

Deceleration distance of 650 m at landing with a weight of $w_L = 0.75 \cdot w_{TO}$ at an airport at mean sea level.

Equation to be used (derivation see lecture notes):

$$\frac{w_{TO}}{S} \leq \frac{s_2 \cdot \bar{a} \cdot C_{L,max,L} \cdot \rho}{1.3225 \cdot \frac{w_L}{w_{TO}}}$$

Coefficients of equation:

- $s_2 = 650 \text{ m}$
- $\bar{a} = a = 3.5 \text{ m/s}^2$ (no thrust reverser installed)
- $C_{L,max,L} = 2.4$
- $\rho = 1.225 \text{ kg/m}^3$ (mean sea level)
- $\frac{w_L}{w_{TO}} = 0.75$

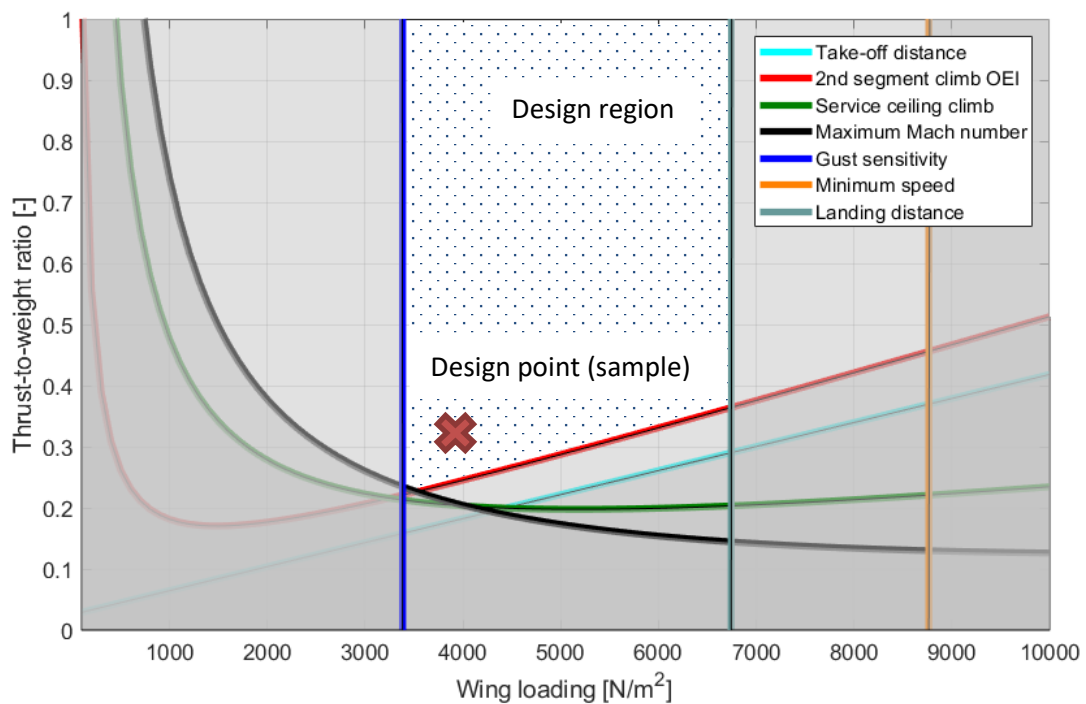
Therefore:

$$\frac{w_{TO}}{S} \leq \frac{650 \cdot 3.5 \cdot 2.4 \cdot 1.225}{1.3225 \cdot 0.75} = 6740 \text{ N/m}^2$$

Task 8

Finally, suggest a design point, which fulfills the above requirements.

Drawing of the design chart with all constraint curves as derived above:



Design point selection:

- Always include a safety margin for design changes (remember that the design chart is only valid for one configuration set!).
- The more to the right, the smaller the wing reference area (assuming constant mass).
- The more to the bottom the smaller the required thrust and therefore the engine (assuming constant mass).
- Priority 1: low engine weight (=low T/W), priority 2: lowest reasonable(!) wing area (= high w/S) → this is a trade-off
- Caution with C_L -variations along the curves (e.g. for OEI the more to the right, the higher is C_L) → make sure the high lift system is able to provide this.